

LEARN
UNDERSTAND
REVISE
SUCCEED

CURRENT AFFAIRS
STATIC LINKAGES
EXAM FOCUS
VISUAL LEARNING

UPSC

CURRENT AFFAIRS

THROUGH

INFOGRAPHICS

IMPORTANT ISSUES | KEY CONCEPTS | EXAM-ORIENTED INSIGHTS



CONCISE



VISUAL



EXAM FOCUSED



RELIABLE



EASY REVISION

UPDATED
FOR
2026



SCO BISHKEK DECLARATION

REGIONAL SECURITY
& GLOBAL ORDER



IST RULES 2026

TIME FOR A
DIGITAL INDIA



INDIA-BELGIUM RELATIONS

TRADE, TECHNOLOGY
AND A STRONGER PARTNERSHIP



SEMICON INDIA 2026

SILICON TO SYSTEMS:
BUILDING THE ECOSYSTEM



MARINE FISHERIES & INDIA'S BLUE ECONOMY

HEALTHY OCEANS
PROSPEROUS PEOPLE



AI vs COPYRIGHT LAW

INNOVATION, CREATORS' RIGHTS
AND THE ROAD AHEAD



AND MORE
KEY DEVELOPMENTS
SHAPING INDIA
AND THE WORLD



A WELL-INFORMED
ASPIRANT BUILDS
A STRONGER INDIA



ECONOMY



INTERNATIONAL
RELATIONS



SCIENCE &
TECHNOLOGY



ENVIRONMENT



POLITY &
GOVERNANCE



SOCIETY



DEVELOPMENT

FOR UPSC
AND OTHER
COMPETITIVE EXAMS

KNOWLEDGE TODAY A STRONGER TOMORROW

ISRO LAUNCHES GSLV-F17 / EOS-05

Strengthening India's Earth Observation Capabilities



SPACE FOR
A SAFER, STRONGER
AND A BRIGHTER INDIA

Launch Date **4 September 2026**

Launch Vehicle **GSLV-F17**
(19th flight of GSLV)

Launch Site Satish Dhawan Space Centre,
Sriharikota

Payload **EOS-05**
(Earth Observation Satellite)

Mission Status **Successfully launched**



EOS-05

An advanced Earth observation satellite to provide high-resolution images for a wide range of applications.

ABOUT EOS-05

- Type** Earth Observation Satellite
- Objective** High-resolution imaging of the Earth's surface
- Applications**
 - Agriculture and crop monitoring
 - Disaster management
 - Forest and environmental monitoring
 - Urban and infrastructure planning
 - Cartography and resource mapping
- Builds on** Success of the EOS series (e.g., Cartosat, Resourcesat)

GSLV-F17: KEY FEATURES

- Three-stage launch vehicle** Solid + Liquid + Cryogenic
- Equipped with an **indigenous cryogenic engine** (CE-7.5)
- 19th flight of GSLV**
- Designed to place heavier satellites in higher orbits
- Demonstrates India's maturity in cryogenic technology

WHY IT MATTERS?

- Enhances India's Earth observation capabilities
- Supports disaster preparedness and climate action
- Aids in sustainable development and resource management
- Strengthens India's self-reliance in space technology
- Contributes to national security and strategic planning

COMPARISON: PSLV vs GSLV vs LVM3

Feature	PSLV	GSLV (Mk II)	LVM3 (Mk III)
Full Form	Polar Satellite Launch Vehicle	Geosynchronous Satellite Launch Vehicle	Launch Vehicle Mark 3
Configuration	4-stage (S-L-S-L)	3-stage (S-L-C)	3-stage (S-L-C) with strap-ons
Payload Capacity	~1,750 kg to LEO	~2,500 kg to GTO	~4,000 kg to GTO ~8,000 kg to LEO
Typical Use	Small and medium satellites, Earth observation	Heavier satellites, communication satellites	Very heavy satellites, human spaceflight (Gaganyaan)
Key Highlights	Workhorse of ISRO, high reliability	Uses cryogenic engine (CE-7.5)	Most powerful Indian launch vehicle, enables big missions

WHAT IS A CRYOGENIC ENGINE?



- Uses liquid hydrogen (fuel) and liquid oxygen (oxidiser) at extremely low temperatures.
- Provides high thrust and efficiency.
- Essential for placing heavy satellites in higher orbits (GTO).
- Indigenous cryogenic engine (CE-7.5) marks a major technological milestone for India.

TYPES OF ORBITS

POLAR ORBIT (LEO)



- Low Earth Orbit (LEO) ~500-800 km
- Passes over the poles
- Useful for Earth observation, mapping, surveillance
- Examples: Cartosat, Resourcesat, EOS series

GEOSTATIONARY ORBIT (GEO)



- ~35,786 km above Earth
- Hovers over the equator
- Appears fixed over a point on Earth
- Used for communication, weather satellites
- Examples: INSAT, GSAT series

THE BIGGER PICTURE

- Strengthens India's position as a leading space power
- Supports Atmanirbhar Bharat in high-technology sectors
- Enables data-driven governance and development
- Contributes to global efforts on climate change, disaster resilience and sustainable future

INDIA REACHES HIGHER

VIKSIT BHARAT
THROUGH SPACE

INDIA'S GDP GROWS 7.8% IN Q1 FY27

Strong domestic demand and record exports support growth

GS-III | ECONOMY

FOR A STRONGER INDIA

HIGHER
GROWTH
STRONGER
INDIA



Real GDP Growth
(Q1 FY27)

7.8%

(Apr–Jun 2026)



India's Goods Exports
(June 2026)

\$44.24 billion

(Record high)



Key Driver

Strong exports,
resilient domestic
demand



Challenge

Lower-margin /
labour-intensive
exports under
pressure

WHAT IS GDP?

Gross Domestic Product (GDP) is the total monetary value of all final goods and services produced within a country's borders in a given period of time.



Goods
(e.g., food, cars)



Services
(e.g., banking, IT)



Only final
output is counted
(to avoid double
counting)

GDP vs GVA

GDP
(Gross Domestic
Product)

- Value of all final goods and services
- Includes indirect taxes minus subsidies

GVA
(Gross Value Added)

- Value added at each stage of production
- Does not include indirect taxes and subsidies

Relationship: $GDP = GVA + \text{Indirect Taxes} - \text{Subsidies}$

REAL vs NOMINAL GDP

Nominal GDP



- Measured at current market prices
- Affected by inflation

Real GDP



- Measured at constant (base year) prices
- Shows actual growth in physical output
- Q1 FY27 growth (7.8%) is real GDP growth

TRENDS IN INDIA'S GDP GROWTH

Real GDP Growth Rate (YoY)



HOW GDP IS CALCULATED? (Expenditure Method)

$$GDP = C + I + G + (X - M)$$

- C** Private Final Consumption Expenditure (Households)
- I** Gross Fixed Capital Formation (Investment)
- G** Government Final Consumption Expenditure
- X** Exports of Goods and Services
- M** Imports of Goods and Services (Net exports = $X - M$)

EXPORTS AND GDP



India's goods exports rose to \$44.24 billion in June 2026 (record high).



Higher exports increase net exports ($X - M$), directly adding to GDP.



Key sectors: engineering goods, petroleum products, electronics, pharmaceuticals, chemicals.



However, lower-margin and labour-intensive exports face global headwinds (rising competition, trade barriers, slower demand).

WHY GROWTH MATTERS?



More jobs and higher incomes



Supports manufacturing
and services expansion



Increases government revenue
for welfare and infrastructure



Helps achieve Viksit Bharat @ 2047

FACTORS SUPPORTING Q1 FY27 GROWTH

- ✓ Strong goods exports (record \$44.24 billion in June)
- ✓ Resilient domestic consumption
- ✓ Continued investment activity
- ✓ Government capital expenditure
- ✓ Stable macroeconomic fundamentals

KEY CHALLENGES AHEAD

- ⚠ Global economic uncertainty and trade barriers
- ⚠ Pressure on labour-intensive exports
- ⚠ Commodity price volatility
- ⚠ Need to sustain investment and job creation

KEY TAKEAWAYS

- 1 India's real GDP grew 7.8% in Q1 FY27.
- 2 Exports are a major growth driver, with goods exports at \$44.24 billion in June.
- 3 Understand GDP vs GVA, real vs nominal GDP and the expenditure method.
- 4 High growth supports jobs, investments and inclusive development, but external risks remain.
- 5 Sustained reforms and global competitiveness are key for long-term growth.



SCO BISHKEK DECLARATION

Stronger Cooperation for a Stable, Secure
and Prosperous Region

SCO SUMMIT 2026
BISHKEK, KYRGYZSTAN

Dialogue | Security | Development

KEY HIGHLIGHTS OF THE BISHKEK DECLARATION



Strong opposition to terrorism, separatism and extremism.



Rejection of "double standards" in counter-terrorism efforts.



Emphasis on dialogue and diplomacy for resolving conflicts.



Commitment to regional stability, connectivity and economic cooperation.

FROM SHANGHAI FIVE TO SCO

- **1996:** Shanghai Five formed by China, Russia, Kazakhstan, Kyrgyzstan and Tajikistan to build trust and reduce border tensions.
- **2001:** Uzbekistan joined and the grouping was formally transformed into the Shanghai Cooperation Organisation (SCO).
- Gradual expansion with new members, observers and dialogue partners.

ABOUT SCO

- Established:** 2001 (evolved from Shanghai Five, 1996)
- Objective:** Promote regional security, stability, economic cooperation and cultural ties.
- Nature:** A Eurasian political, economic and security grouping.
- Secretariat:** Beijing, China
- Current Chair:** Kyrgyzstan (2026)
- Next Chair:** Tajikistan (2027)
- Working Mechanism:** Annual summits, Council of Heads of State, Council of Ministers, RATS, sectoral meetings.

MEMBERS, OBSERVERS & DIALOGUE PARTNERS

Members (10)



Observer States (2)



Dialogue Partners (14)



RATS – REGIONAL ANTI-TERRORIST STRUCTURE

- **Established:** 2004 (Tashkent, Uzbekistan)
- **Function:** Coordinate efforts against terrorism, separatism and extremism.
- **Activities:** Intelligence sharing, capacity building, joint exercises, database of terrorist organisations.
- **Significance:** Institutional mechanism to address common security threats in the region.

INDIA AND SCO

- Joined as a full member in 2017** (along with Pakistan).
- Actively participates in counter-terrorism, connectivity, economic and cultural initiatives.**
- At the Bishkek Summit, India reiterated dialogue and diplomacy for resolving conflicts and highlighted the need for a rules-based, multilateral order.**
- Supports a united stance against terrorism and rejects double standards.**
- Uses SCO as a platform to engage with Central Asian states and Eurasia.**

INDIA'S CENTRAL ASIA POLICY

- Strengthen historical and cultural ties with Central Asian countries.**
- Expand cooperation in trade, connectivity, energy, defence, critical minerals and education.**
- Key initiatives:** International North-South Transport Corridor (INSTC), Chabahar Port, Central Asia-India Dialogue, development partnerships.
- Important for energy security, access to resources and balancing regional powers.**
- Part of India's broader "Extended Neighbourhood" and Act East-Connect Central Asia approach.**

CHINA/RUSSIA FACTOR

- China**
 - China seeks to use SCO to expand its influence and advance BRI.
 - India remains cautious about China's strategic assertiveness.
- Russia**
 - Russia views SCO as a key platform to maintain its influence in Eurasia.
 - India continues to engage with Russia while balancing its ties with other partners.

INDIA'S STRATEGIC INTERESTS IN SCO

- ✓ Counter-terrorism cooperation.
- ✓ Access to Central Asian markets and resources.
- ✓ Connectivity (INSTC, Chabahar, regional corridors).
- ✓ Balancing China's influence.
- ✓ Energy security and critical minerals.
- ✓ Supporting a multipolar, rules-based international order.



PEACEFUL NEIGHBOURHOOD
PROSPEROUS REGION
STRONGER INDIA

COOPERATION TODAY
A MORE SECURE TOMORROW

INDIAN STANDARD TIME (IST) RULES 2026

A common, legally enforceable time reference for a digitally connected India



ONE TIME
ONE INDIA



ACCURACY
RELIABILITY



SECURITY
RESILIENCE



ENABLING
DEVELOPMENT

IST
UTC + 5:30

India's Official
Time Reference

SYNCHRONISED
TIME
STRONGER INDIA

CSIR-NPL



WHAT ARE THE IST RULES 2026?



- A new legal and technical framework to define, maintain and disseminate Indian Standard Time (IST).
- Provides a common, accurate and enforceable national time reference for all sectors.
- Ensures synchronisation of time across government, industry and citizens.
- Brings timekeeping under a robust regulatory and technical system.

KEY PROVISIONS



Legally enforceable
Indian Standard Time



Authorises CSIR-NPL as the
national timekeeper



Standardised time dissemination
across telecom, internet, broadcast
and critical infrastructure



Periodical calibration and
audit mechanisms



Defines responsibilities for
government and private entities



Measures for security, redundancy
and resilience

WHO MAINTAINS IST?



वेद्यमित्र सेवा राष्ट्र सेवा

CSIR-NPL
National Physical
Laboratory, New Delhi

- India's apex institute for timekeeping.
- Maintains IST using advanced atomic clocks.
- Generates and disseminates time signals through multiple channels.
- Works under the Council of Scientific and Industrial Research (CSIR), Ministry of Science & Technology.

**CSIR-NPL is India's official source
for Indian Standard Time.**

HOW IS IST GENERATED?



ATOMIC CLOCKS

Ultra-precise cesium
atomic clocks at CSIR-NPL
measure time.



UTC COMPARISON

IST is kept in sync with
Coordinated Universal Time (UTC)
with an offset of +5:30 hours.



DISSEMINATION

Time signals are shared via
telecom networks, internet (NTP),
broadcast, and dedicated services
to government and industry.

UNDERSTANDING THE TIME TERMS



UTC
(Coordinated
Universal Time)

- Global time standard maintained by BIPM.
- Based on atomic clocks.
- Not affected by time zones.



IST
(Indian Standard Time)

- UTC + 5:30 hours
(5 hours 30 minutes ahead).
- Single time zone for the
entire country.
- No daylight saving time.



**ATOMIC
CLOCKS**

- Use the natural vibration
of atoms (usually cesium).
- Extremely accurate (error
of less than a second
in millions of years).
- Foundation for UTC and IST.

WHY SYNCHRONISED TIME MATTERS?



Digital Infrastructure

Essential for telecom, internet, 5G/6G,
cloud services and data centres.



Cybersecurity

Prevents time spoofing; critical for
encryption, digital signatures and secure
transactions.



Financial Systems

Accurate time is vital for stock markets,
banking and digital payments.



Transport & Logistics

Navigation, railways, aviation and shipping
depend on precise time.



Power Grids

Synchronisation ensures stable and reliable
electricity supply.



Governance & Public Services

Supports e-governance, emergency services
and legal processes.



Space & Defence

Critical for satellite operations, navigation
(GPS/NavIC) and secure communications.

LEGAL METROLOGY PERSPECTIVE



- Time measurement falls under the Legal Metrology Act, 2009.
- The IST Rules 2026 provide the legal basis for enforcement, compliance and penalties in case of tampering or misuse.
- Ensures uniformity across industries and service providers.

KEY TAKEAWAYS

- 1 IST Rules 2026 provide a common and enforceable national time reference.
- 2 CSIR-NPL maintains IST using atomic clocks.
- 3 IST = UTC + 5:30 (no daylight saving time).
- 4 Accurate time is critical for cybersecurity, finance, transport, power and digital India.
- 5 Strengthens trust, security and resilience in a connected economy.

THE BIGGER PICTURE



- ✓ A synchronised India
- ✓ A secure digital future
- ✓ Stronger economy
- ✓ Reliable infrastructure
- ✓ Greater national security

ACCURATE TIME
CONNECTS PEOPLE
POWERS PROGRESS

SAME TIME
A STRONGER TOMORROW

INDIA-BELGIUM RELATIONS **UPGRADED**

Stronger Partnerships for a Shared Prosperity

DEMOCRACIES
PARTNERS
FOR A BETTER
TOMORROW



**STRONGER
PEOPLE-TO-PEOPLE TIES**



**CLEANER
GREENER FUTURE**



**INNOVATION
& TECHNOLOGY**



**SHARED VALUES
A STRONGER WORLD**

Grand Place, Brussels
(Belgium)

KEY ANNOUNCEMENTS



India and Belgium agreed to deepen cooperation in trade, defence, clean energy, critical minerals, semiconductors, research and talent mobility.



Investment Fast-Track Mechanism
to facilitate and accelerate investments.



**Goal to double bilateral trade
within five years.**



Enhanced collaboration in education, research, innovation and talent mobility.



Commitment to a long-term, strategic partnership based on shared democratic values.

AREAS OF COOPERATION



**Trade &
Investment**



**Defence &
Security**



**Clean Energy
& Climate Action**



**Critical
Minerals**



**Semiconductors
& Advanced
Manufacturing**



**Research, Innovation
& Technology**



**Education &
Talent Mobility**



**People-to-People
Ties**



**Sustainable
Development**

WHY BELGIUM MATTERS?



Strategic location

Gateway to the European Union, hosts EU institutions (EU Commission and EU Council).



Economic strength

Highly developed economy with strengths in chemicals, pharmaceuticals, engineering, logistics and high-tech industries.



Innovation hub

Home to world-class research institutions, universities and R&D ecosystem.



Key role in global supply chains

Important player in semiconductors, pharmaceuticals and critical raw materials.



Influence in global governance

Active in multilateral forums, supports a rules-based international order.

INDIA-EU RELATIONS CONTEXT



- Belgium is an important partner within the European Union.
- India-EU Strategic Partnership covers trade, connectivity, climate change, clean energy, digital technology and people-to-people ties.
- Belgium often supports closer India-EU economic and political engagement.
- Forthcoming India-EU Free Trade Agreement (FTA) will further strengthen ties.

**Stronger India-Belgium ties
contribute to a deeper and broader
India-EU partnership.**

SEMICONDUCTORS



- Belgium is a key player in the global semiconductor supply chain (e.g., advanced materials, equipment, research).
- Cooperation can support India's semiconductor mission through technology, R&D and skilled talent.
- Reduces dependence on limited geographies and strengthens supply-chain resilience.
- Opens opportunities for joint research, manufacturing and innovation.

CRITICAL MINERALS



- Belgium plays an important role in trading and processing critical minerals.
- Cooperation ensures secure, diversified and sustainable supply chains for India's clean energy transition.
- Helps India's green hydrogen, electric vehicles, renewable energy and high-tech manufacturing sectors.
- Aligns with shared goal of building resilient and responsible mineral supply chains.

BILATERAL TRADE



Current trade (2024-25)
~ US\$ 13.5 billion
(approx.)



Target
**Double trade
within 5 years**

Major items of trade



India's exports

Pharmaceuticals, chemicals, textiles, machinery, agri-products, IT services



India's imports

Diamonds, machinery, chemicals, electronic goods, metals and high-tech products

KEY TAKEAWAYS

- 1 Upgradation reflects growing trust and convergence of interests.
- 2 Opens new opportunities in trade, defence, clean energy, critical minerals and semiconductors.
- 3 Supports India's economic growth, technological advancement and green transition.
- 4 Strengthens India-EU partnership and India's global standing.
- 5 A win-win partnership for a stable, prosperous and sustainable future.



INDIA BELGIUM

DIFFERENT
HORIZONS
COMMON
OPPORTUNITIES

Atomium, Brussels
(Symbol of innovation and progress)



PEOPLE | PARTNERSHIP | PROGRESS

**STRONGER TODAY
BRIGHTER TOMORROW**

SEMICON INDIA 2026

TECHNOLOGY
INNOVATION
ATMANIRBHAR BHARAT

Silicon to Systems: Building the Ecosystem

India's platform to power a self-reliant and globally connected semiconductor ecosystem



Dates
17 – 19
September 2026



Venue
New Delhi
(India)



Edition
5th
Edition



Theme
Silicon to Systems:
Building the Ecosystem

ABOUT SEMICON INDIA



Flagship event bringing together government, industry, investors, innovators and global partners.



Showcases opportunities across the semiconductor value chain.



Platform for policy dialogue, partnerships, technology showcase and investment promotion.



Positions India as a key player in the global semiconductor ecosystem.

KEY FOCUS AREAS (2026)



Manufacturing and supply chains



ATMP/OSAT and design ecosystem



Emerging technologies (AI, EV, IoT, 5G)



Global partnerships and investments



Startup innovation and talent development



Resilient and secure supply chains



Sustainability and green semiconductors

ORGANISED BY



Ministry of Electronics
and Information Technology
Government of India



semicon india

BUILDING A RESILIENT SEMICONDUCTOR ECOSYSTEM

Towards a Viksit Bharat
through Semiconductors

INDIA SEMICONDUCTOR MISSION (ISM)



INDIA SEMICONDUCTOR MISSION

Catalyzing India's Semiconductor Ecosystem

- Launched in 2021 under MeitY.
- Objective: Develop a vibrant semiconductor and display manufacturing ecosystem in India.
- Financial support for fabs, ATMP/OSAT units, design infrastructure, and R&D.
- Focus on creating a self-reliant, innovation-driven and globally competitive semiconductor sector.
- Total outlay: ~₹76,000 crore.

SEMICON 2.0 (NEXT PHASE)

- Builds on the momentum of ISM.
- Focus on larger, more advanced fabs.
- Greater support for design, R&D and compound semiconductors (SiC, GaN).
- Incentives for equipment & materials manufacturing.
- Skilling and talent development at scale.
- Stronger integration with global supply chains.
- Aims to position India as a global semiconductor hub.

KEY SEGMENTS OF THE VALUE CHAIN



Fab
(Wafer
Fabrication)

Manufactures semiconductor wafers (front-end processes).



ATMP/OSAT
(Assembly, Testing,
Marking & Packaging)

Packages and tests chips (back-end processes).



Design

Chip design, IP cores and EDA tools.

Complete ecosystem: Materials + Equipment + Design + Manufacturing + Packaging + Testing + End-use industries

SUPPLY-CHAIN SECURITY



Reduces dependence on a single country for critical chips.



Diversifies sources of raw materials, equipment and technology.



Builds domestic manufacturing and trusted supply chains.



Enhances resilience against geopolitical shocks and disruptions.

CHINA-US COMPETITION: IMPLICATIONS



China

- Dominates global chip manufacturing (high share in mature nodes).
- Strong control over critical minerals and supply chains.
- US export restrictions on advanced chips and equipment.
- India seeks to reduce over-dependence on China.



United States

- Leads in advanced chip design, technology and equipment.
- Imposes export controls to limit China's access to high-end chips.
- Seeks trusted partners to diversify supply chains ("China+1").
- India is seen as a key partner in building resilient supply chains.

WHY IT MATTERS FOR INDIA?



Supports Atmanirbhar Bharat and economic growth.



Generates high-skilled jobs.



Enables domestic manufacturing in strategic sectors (electronics, defence, automotive, telecom, AI).



Strengthens technological sovereignty and national security.



Positions India in the evolving global semiconductor value chain.

FROM CHIPS TO A STRONGER INDIA



SELF-RELIANCE



INNOVATION



GLOBAL PARTNERSHIP



SUSTAINABLE GROWTH

SEMICONDUCTORS
POWER
PEOPLE
PROSPERITY



MARINE FISHERIES & INDIA'S BLUE ECONOMY

Healthy Oceans, Prosperous People, Stronger India

CLEAN OCEANS
THRIVING COASTS
VIKSIT BHARAT



FOOD
NUTRITION



MILLIONS OF
LIVELIHOODS



EXPORT
EARNINGS



SUSTAINABLE
OCEAN GROWTH



BLUE ECONOMY
STRONG OCEANS
STRONGER INDIA

INDIA'S MARINE RESOURCES



MARINE FISHERIES AT A GLANCE



4.3 million tonnes

Marine fish production
(2023–24, approx.)



~ 4 million

People depend on marine fisheries (fishers, fish farmers, workers, allied activities)



~ ₹60,000+ crore

Export earnings from marine products (2023–24)



Key to nutrition

Provides high-quality protein, essential omega-3 fatty acids and micronutrients



Major species

Tuna, shrimp, sardine, mackerel, seer fish, pomfret, squid, cuttlefish, lobster, crab etc.

WHAT IS BLUE ECONOMY?

Sustainable use of ocean resources for economic growth, improved livelihoods and ocean ecosystem health.



Fisheries & Aquaculture



Maritime Transport & Ports



Offshore Energy (oil, gas, renewables)



Coastal Tourism



Marine Biodiversity



Blue Technology & Ocean Innovation



Seabed Minerals (Polymetallic nodules, polymetallic sulphides, etc.)

Blue Economy = Economic Growth + Healthy Oceans + Resilient Coastal Communities

PM MATSYA SAMPADA YOJANA (PMMSY)



(2020–21 to 2024–25)



Objective: Sustainable and responsible development of fisheries sector.



Total Investment: ~ ₹20,050 crore (Central share ~ ₹9,407 crore)



Key Components:

- Enhancing fish production and productivity
- Infrastructure (fishing harbours, landing centres, cold chain)
- Fisheries value chain and marketing
- Aquaculture development
- Fishers' welfare and social security
- Monitoring, control and surveillance (against IUU fishing)

SUSTAINABLE FISHING & CONSERVATION



Prevent overfishing

Follow science-based catch limits and seasonal bans (e.g., breeding season).



Reduce bycatch

Use selective and eco-friendly fishing gear.



Protect marine ecosystems

Conserve coral reefs, mangroves, seagrass and breeding grounds.



Combat IUU fishing

Strengthen monitoring, control and surveillance (MCS) using technology (VMS, AIS, drones).



Promote responsible aquaculture

Adopt low-impact, climate-resilient and sustainable practices.

UNCLOS & INDIA



• UNCLOS (1982) is the legal framework governing oceans and their resources.

• Defines maritime zones:

- Territorial Sea – up to 12 nm
- Contiguous Zone – up to 24 nm
- Exclusive Economic Zone (EEZ) – up to 200 nm
- Continental Shelf – beyond 200 nm (up to 350 nm in some cases)
- Gives sovereign rights over resource exploration and exploitation in the EEZ and continental shelf.
- India is a party to UNCLOS and actively participates in its institutions (e.g., ISA – International Seabed Authority).

UNCLOS enables India to harness its ocean wealth while ensuring responsible and sustainable use.



CHALLENGES



Overfishing and declining fish stocks



Climate change (ocean warming, acidification)



Marine pollution (plastic, oil spills, ghost nets)



IUU fishing by foreign vessels



Habitat loss (mangroves, coral reefs)



Livelihood vulnerability of coastal communities



Need for better data, research and enforcement



OPPORTUNITIES



Rising global demand for seafood



Value addition and export diversification



Deep-sea fishing and under-exploited resources (e.g., tuna)



Mariculture and seaweed cultivation



Ocean-based renewable energy (offshore wind, tidal)



Blue biotechnology and pharmaceuticals



Coastal and island tourism



Employment generation and women's participation



WAY FORWARD



Implement science-based fisheries management



Strengthen MCS and curb IUU fishing



Invest in sustainable aquaculture and eco-friendly infrastructure



Empower fishers with technology, credit and social security



Enhance research in marine biodiversity and climate resilience



Regional and global cooperation for healthy oceans



Mainstream Blue Economy in national growth strategy

**“Healthy Oceans Today
for Prosperous Generations Tomorrow”**

VIKSIT BHARAT 2047



SUSTAINABLE
RESOURCES



RESILIENT
COMMUNITIES



THRIVING
ECONOMY



CLEANER
OCEANS

AI vs COPYRIGHT LAW

Innovation, Intellectual Property and a Fair Future

How do we balance the promise of generative AI with the rights of creators?

CREATIVITY FUELS HUMAN PROGRESS. LET IT THRIVE IN THE AI ERA.

TEXT
IMAGES
MUSIC
VIDEO
BOOKS
NEWS
CODE
IDEAS

GS-II / GS-III
UPSC CURRENT AFFAIRS

THE ISSUE

- Generative AI models are trained on massive amounts of data, often including copyrighted material (text, images, music, etc.).
- Creators and publishers allege unauthorised use of their works.
- AI companies argue it is necessary for innovation and may be covered under exceptions like fair dealing (fair use).
- The debate is about how to balance creators' rights with technological progress.

KEY STAKEHOLDERS

- Creators** (authors, artists, musicians, filmmakers, publishers)
Seek recognition, consent and fair compensation.
- AI companies**
Need large datasets to train models; argue for flexibility and legal clarity.
- Users / Public**
Access to innovative and affordable AI products.
- Governments & Regulators**
Balance innovation, IP rights and public interest.

RELEVANT LEGAL FRAMEWORK

- Copyright Act, 1957 (India)**
 - Protects original literary, dramatic, musical and artistic works, etc.
 - Rights include reproduction, communication to the public, adaptation and translation.
 - Duration: Generally life of author + 60 years.
- Key Provisions / Concepts**
 - Section 14** – exclusive rights of copyright owner.
 - Section 51** – what constitutes infringement.
 - Section 52** – exceptions (fair dealing for research, private use, criticism, review, reporting of current events, etc.).
 - Section 57** – moral rights (right to attribution, integrity).

FAIR DEALING IN THE AI CONTEXT

- AI training may be argued as:
 - "Fair dealing" for research or non-commercial purposes.
 - "Transformative use" (creates new outputs, not a substitute).
 - Concerns:
 - Large-scale, commercial use may go beyond fair dealing.
 - Exact reproduction or close outputs can harm the market for original works.
- Whether AI training is 'fair dealing' is a key legal question, and courts worldwide are examining it.**

KEY CONCERNS

- Unauthorised use of copyrighted works for training data.
- Loss of income for creators and industries (music, publishing, media).
- Risk of plagiarism and generation of similar/competing works.
- Lack of transparency about training data.
- Bias and misuse of copyrighted or sensitive content.
- Impact on cultural diversity and traditional knowledge.

GLOBAL DEVELOPMENTS

- United States**
 - Courts examining 'fair use' in AI training (e.g., lawsuits by authors, news organisations, artists).
 - No specific AI copyright law yet.
- European Union**
 - EU AI Act (2024) requires transparency about training data and respect for copyright.
 - Text and Data Mining (TDM) exceptions with opt-out for rightsholders.
- United Kingdom**
 - Consultations on copyright and AI.
 - Proposes a more flexible regime with transparency requirements.
- China**
 - Guidelines recognise copyright protection, with some allowances for training under specific conditions.

INDIA'S APPROACH

- No specific AI law yet; existing Copyright Act, 1957 applies.
- Government and expert bodies are examining the issue (consultations, committee discussions).
- Focus areas:
 - Need for transparency in training data.
 - Possible licensing or collective management systems.
 - Protection of creators' rights while enabling innovation and AI for public good.
- Also linked to India's digital economy, AI strategy and creative industry growth.

India needs a balanced, future-ready framework that protects creators, promotes innovation and leverages AI for inclusive growth.

POSSIBLE SOLUTIONS

- Licensing models**
Structured access to copyrighted content.
- Collective management organisations**
Pooling of rights and revenue sharing.
- Transparency requirements**
Disclosure of training data sources.
- Compensation mechanisms**
Revenue sharing or data compensation funds.
- Clear guidelines on fair dealing**
Define permissible uses for AI training.
- Promote open datasets**
Publicly available, licensed content.

IMPLICATIONS FOR UPSC

- GS-II (Polity & Governance)**
 - Intellectual property rights
 - Regulation of emerging technologies
 - Role of government in balancing innovation and rights
- GS-III (Science & Technology)**
 - AI and its applications
 - Impact on economy and creative industries
 - Ethical and legal challenges in technology
- Essay / Ethics / Interdisciplinary**
 - Innovation vs. rights
 - Technology and human values
 - Inclusive growth in the digital age

KEY TAKEAWAYS

- Generative AI has intensified the debate on copyright, with no easy answers.
- Copyright Act, 1957 provides protection, but its application to AI training is evolving.
- Global approaches vary — from flexible exceptions to stricter transparency norms.
- India needs a balanced framework that ensures creator compensation, encourages innovation and aligns with global best practices.
- The issue has far-reaching implications for the creative economy, digital rights and the future of AI governance.

SUPPORT CREATIVITY. ENABLE INNOVATION. BUILD A FAIR AI FUTURE.



DAILY CURRENT AFFAIRS

KEY NEWS | DEEPER INSIGHTS | EXAM FOCUS

6 IMPORTANT TOPICS TODAY

5 SEPTEMBER 2026

1 INDIA'S FOREX RESERVES HIT A FRESH RECORD



RBI data show foreign-exchange reserves reach **\$740.8 billion** for the week ended August 28.

GS-III | ECONOMY

2 RBI MOVES TO ABSORB RECORD LIQUIDITY



₹7-trillion, 30-day VRRR auction announced against a banking-system liquidity surplus of around ₹10.3 trillion.

GS-III | ECONOMY

3 INDIA'S STRONG GDP GROWTH SPARKS A DEBATE



High growth numbers, but questions remain on employment and investment.

GS-III | ECONOMY

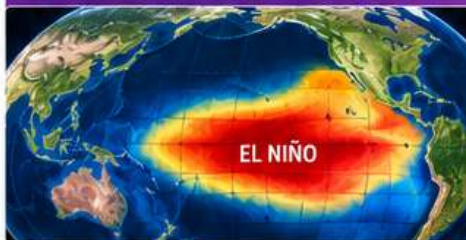
4 BELOW-NORMAL SEPTEMBER RAINFALL RAISES CROP CONCERNS



Uneven monsoon and below-normal September rainfall may affect kharif output and food inflation.

GS-I | GEOGRAPHY | GS-III

5 SUPER EL NIÑO RISK



Possibility of an unusually strong El Niño event and its wide-ranging impacts on weather and the Indian monsoon.

GS-I | GEOGRAPHY | GS-III

6 NASA'S NANCY GRACE ROMAN SPACE TELESCOPE



A next-generation observatory to study dark energy, dark matter, exoplanets and more.

GS-III | SCIENCE & TECHNOLOGY



KNOW
ANALYSE
APPLY
EXCEL



CLEAN FACTS



CONCEPT
CLARITY



EXAM ORIENTATION



A BRIGHTER TOMORROW

**CURRENT AFFAIRS
TODAY**
A STRONGER YOU TOMORROW

SONE RIVER WATER DISPUTE

RIVERS CONNECT PEOPLE.
FAIR SHARING BUILDS A STRONGER INDIA.

A DISPUTE
OVER WATER,
A SHARED
RESPONSIBILITY.

COOPERATIVE
FEDERALISM
FOR A WATER
SECURE INDIA



Sone Barrage, Dehri (Bihar)

ABOUT THE SONE (SON) RIVER



Also known as: Son River (Sone)



Origin: Amarkantak Plateau,
Maikal Hills (Madhya Pradesh)



Length: ~784 km



Major tributaries: Rihand, North Koel,
Johilla, Gopad, Kanhar



Flows through: Madhya Pradesh →
Uttar Pradesh → Bihar



Confluence: Joins the Ganga near
Patna (Bihar)



Part of: Ganga river basin



Importance: Irrigation, drinking water,
hydropower, and supports agriculture
in the eastern Indo-Gangetic plains.

COURSE OF THE SONE RIVER



WHY IS THERE A DISPUTE?



Sharing of water

Differing state interests on allocation
for irrigation, power and drinking water.



Upstream vs downstream concerns

Upstream projects may reduce water
availability downstream.



Irrigation demands

High demand for irrigation in
water-scarce regions (especially during
lean season).



Multipurpose projects

Proposals for dams and barrages
(e.g., Kanhar, Bansagar, etc.) have
raised concerns.



Lack of a final settlement

Demands for a tribunal and formal
allocation persist.

KEY PROJECTS ON THE SONE RIVER

Project	Location	Purpose	Key points
Indrapuri Barrage	Dehri (Bihar)	Irrigation, water supply, power	Major existing structure (commissioned in 1960s).
Bansagar Project	Madhya Pradesh (U.P.-MP border)	Irrigation, power	Multipurpose project; water sharing concerns.
North Koel Project	Jharkhand	Irrigation	Proposed; linked to inter-state discussions (with Koel and Son basins).
Kanhar Project	Madhya Pradesh	Irrigation	Proposed; part of Madhya Pradesh's plans.
Rihand Dam	U.P. (on tributary)	Irrigation, hydropower	On major tributary of Son.

LEGAL AND INSTITUTIONAL FRAMEWORK



Article 262 of the Constitution

- Empowers Parliament to provide for adjudication of inter-state river water disputes.
- Parliament can also bar the jurisdiction of the Supreme Court in such disputes.



Inter-State River Water Disputes Act, 1956

- Provides for the constitution of a tribunal to adjudicate disputes.
- Tribunal's decision is final and binding on the parties.



Inter-State River Water Disputes (Amendment) Act, 2002

- Streamlined the process for constitution of tribunals.
- Provides for a permanent tribunal mechanism (now through the Inter-State River Water Disputes Tribunal).

SONE RIVER DISPUTE: KEY STAKEHOLDERS



Madhya Pradesh

- Proposes projects for irrigation and power.
- Seeks greater share for development needs.



Uttar Pradesh

- Depends on water for irrigation in eastern U.P.
- Concerns over upstream diversions.



Bihar

- Relies on downstream flows.
- Concerns over reduced water availability due to upstream projects.

Need for consultation, data sharing and mutually acceptable agreements.

SIGNIFICANCE OF RESOLVING THE DISPUTE



Improved agricultural productivity and food security



Drinking water security for millions



Greater power generation potential



Balanced regional development



Strengthens cooperative federalism



Supports climate resilience and sustainable water management

CHALLENGES



Divergent state interests



Variable and seasonal river flows



Impact of climate change



Lack of real-time data sharing



Political sensitivities



Delays in constituting tribunals and implementing decisions

THE WAY FORWARD



Early constitution of a tribunal (if needed)



Data-driven and transparent decision-making



Basin-level planning and integrated water resource management



Focus on equitable and sustainable use



Strengthen inter-state consultation mechanisms



Align with the National Water Policy, 2012 and climate adaptation strategies

RELEVANCE FOR UPSC

- GS-II:** Inter-state relations, federalism, dispute resolution mechanisms, role of tribunals, governance.
- GS-I:** River systems of India, Ganga basin, physical geography.
- GS-III:** Water resource management, irrigation, sustainable development.
- Essay:** Cooperative federalism and equitable sharing of natural resources.

KEY TAKEAWAYS

- The Sone River is a vital tributary of the Ganga.
- Disputes arise due to competing demands for water among Madhya Pradesh, Uttar Pradesh and Bihar.
- Article 262 and the Inter-State River Water Disputes Act, 1956 provide the legal framework.
- A cooperative, data-driven and sustainable approach is essential for long-term water security and regional harmony.

**WATER
SHARED WISELY
FUELS A
PROSPEROUS
TOMORROW**

NASA's NANCY GRACE ROMAN SPACE TELESCOPE

A NEW EYE ON THE UNIVERSE

NASA's Nancy Grace Roman Space Telescope (Roman Telescope) will explore the cosmos, from dark energy and dark matter to exoplanets, using a wide field of view and advanced instruments.

Wider. Deeper. More to Discover.
Mapping the Universe. Finding New Worlds.



SAME UNIVERSE. BIGGER QUESTIONS. BOLDER ANSWERS.

WIDE VIEW. DEEP INSIGHTS. BRIGHTER TOMORROW.

KEY FACTS

-  **Agency**
NASA
-  **Launch (planned)**
May 2027
-  **Launch vehicle**
SpaceX Falcon Heavy
-  **Orbit**
L2 (Sun-Earth Lagrange Point)
-  **Mission duration**
At least 5 years (goal: 10 years)
-  **Primary goal**
Wide-field observations to study dark energy, dark matter, exoplanets and the infrared universe

WHO WAS NANCY GRACE ROMAN?

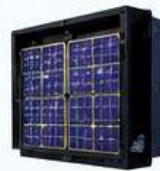


Nancy Grace Roman (1925–2018)

- Pioneering astronomer and NASA's first Chief of Astronomy (1960–1979)
- Played a key role in the Hubble Space Telescope
- Known as the "Mother of Hubble"
- Advocated for large space missions and greater participation of women in science

"The future of astronomy is in space."
– Nancy Grace Roman

MAIN INSTRUMENTS



Wide Field Instrument (WFI)

- 18 detectors, ultra-wide field of view
- 100 times larger area than Hubble's imaging
- High-resolution imaging and slitless spectroscopy
- Will survey billions of galaxies



Coronagraph Instrument (CGI)

- Blocks starlight to detect faint planets
- Will directly image exoplanets
- Demonstrates advanced wavefront-control technology

WHAT WILL ROMAN STUDY?



Dark Energy

Map the large-scale structure of the universe to understand the nature of dark energy and the universe's accelerating expansion.



Dark Matter

Study the distribution of dark matter through gravitational effects (weak lensing).



Exoplanets

Directly image exoplanets and study their atmospheres using the coronagraph.



Gravitational Microlensing

Detect exoplanets, including low-mass and distant planets, using microlensing (gravitational lensing by a foreground star).

HOW IT WILL WORK

1

Wide-field survey

Capture images of large areas of the sky in infrared light.

2

Detect faint signals

Use sensitive instruments to observe distant galaxies, weak gravitational lensing and exoplanets.

3

Data analysis

Map billions of galaxies and detect tiny changes in light caused by dark matter and dark energy.

4

New discoveries

Provide insights into the origins, evolution and future of the universe.

COMPARISON WITH OTHER TELESCOPES

Feature	Hubble	James Webb (JWST)	Roman
Launch year	1990	2021	2027 (planned)
Primary focus	General astronomy	Early universe, IR observations	Wide-field survey, dark energy, exoplanets
Wavelength range	Visible, UV, near-IR	Infrared (0.6 – 28 μm)	Near-IR (0.48 – 2.3 μm)
Field of view	Narrow (~10 arcmin)	Narrow (~9.7 arcmin)	Very wide (~0.28 deg ² ; ~100× Hubble)
Key strength	High-resolution imaging	Deep infrared imaging	Wide-field imaging, cosmic surveys, exoplanet imaging

WHY IS IT IMPORTANT?

-  Will survey billions of galaxies, much larger than previous missions.
-  Helps answer fundamental questions about the nature of dark energy and dark matter.
-  Will expand our search for exoplanets, including Earth-like worlds.
-  Complements Hubble and JWST, providing a wider and deeper view of the universe.

EXPECTED OUTCOMES

-  A detailed map of the universe and its evolution
-  Better understanding of dark energy and dark matter
-  Discovery and characterisation of thousands of exoplanets
-  New insights into galaxy formation, stellar populations and cosmic structure

RELEVANCE FOR UPSC

-  **GS-III: Science & Technology**
 - Space technology and missions
 - Astronomy and astrophysics
 - Applications of space research
-  **Prelims:** Important facts (agency, instruments, L2, objectives, etc.)
-  **Mains:** Significance for scientific advancement, human knowledge and international space collaboration
-  **Also linked to:** International cooperation (in NASA), role of space technology in sustainable development

FROM NEW WORLDS TO BIGGER ANSWERS

THE ROMAN TELESCOPE WILL HELP US SEE FARTHER, UNDERSTAND DEEPER.

"By exploring the universe, we also discover our place in it."

**SCIENCE TODAY
A BRIGHTER TOMORROW**

SUPER EL NIÑO RISK

A STRONGER OCEAN WARMING, BIGGER IMPACTS

Meteorological agencies are warning of the possibility of an unusually strong El Niño event, with wide-ranging implications for global weather, the Indian monsoon, agriculture, marine ecosystems and the world economy.



OCEAN CHANGES SHAPE OUR WEATHER

UNDERSTAND
PREPARE
BUILD RESILIENCE

WHAT IS EL NIÑO?



- El Niño is the warm phase of ENSO (El Niño–Southern Oscillation).
- It refers to the abnormal warming of sea surface temperatures (SST) in the central and eastern equatorial Pacific Ocean.
- Occurs once every 2–7 years, usually lasting 9–12 months.
- Can range from weak to very strong; “Super El Niño” refers to an unusually intense event (e.g., 1982–83, 1997–98, 2015–16).

WHAT IS LA NIÑA?



- La Niña is the cool phase of ENSO.
- Characterised by below-normal SST in the central and eastern Pacific.
- Associated with stronger trade winds and enhanced upwelling of cold water.
- Often leads to above-normal rainfall in many parts of the world, including India (generally).

ENSO CYCLE

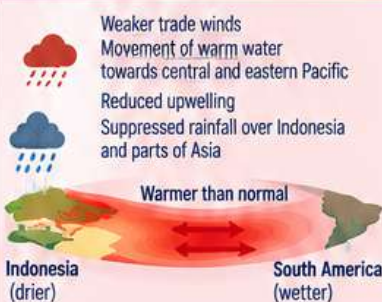


- ENSO is a natural climate phenomenon in the tropical Pacific Ocean.
- It has two opposite phases — El Niño (warm) and La Niña (cool), and a neutral phase.
- Driven by changes in sea surface temperatures, trade winds and ocean–atmosphere interactions.
- Monitored using Oceanic Niño Index (ONI) (3-month average SST anomaly in Niño 3.4 region).

THE WALKER CIRCULATION (NORMAL CONDITIONS)



DURING EL NIÑO



DURING LA NIÑA



CLASSIFICATION (BASED ON ONI)

Category	ONI (°C)
La Niña (weak)	-0.5 to -0.9
La Niña (moderate)	-1.0 to -1.4
La Niña (strong)	≤ -1.5
Neutral	-0.4 to +0.4
El Niño (weak)	+0.5 to +0.9
El Niño (moderate)	+1.0 to +1.4
El Niño (strong)	+1.5 to +1.9
Super El Niño	≥ +2.0

GLOBAL IMPACTS OF A SUPER EL NIÑO

- Higher global temperatures**
Contributes to the warmest years on record.
- Changed rainfall patterns**
Droughts in some regions, floods in others.
- More extreme weather**
Higher frequency and intensity of heatwaves, heavy rainfall and tropical cyclones.
- Food security risks**
Lower crop yields in drought-affected regions.
- Marine ecosystem stress**
Coral bleaching, disruption of fisheries.
- Economic impacts**
Agriculture, fisheries, energy and global trade can be affected.

IMPACT ON INDIA

- Weaker southwest monsoon**
Higher chance of below-normal rainfall (though not always).
- Risk to kharif crops**
Particularly rainfed agriculture (rice, pulses, oilseeds).
- Higher temperatures**
More heatwaves and prolonged dry spells.
- Lower reservoir levels**
Can affect irrigation and drinking water supply.
- Higher food inflation**
Reduced domestic production may increase food prices.
- May influence tropical cyclones**
Shifts in cyclone frequency and tracks in the Indian Ocean region.

HISTORICAL SUPER EL NIÑO EVENTS

Year	Global Impact	Impact on India
1982–83	High global temperatures, widespread droughts	Severe monsoon deficit, drought
1997–98	Major climate disruptions	Weak monsoon, lower agricultural output
2015–16	Warmest year on record (at that time)	Below-normal monsoon, drought in many regions

KEY TAKEAWAYS

- El Niño and La Niña are opposite phases of ENSO.
- Super El Niño (ONI ≥ +2.0°C) is rare but has major global impacts.
- It can weaken the Indian monsoon and affect agriculture, water availability and food prices.
- It influences marine ecosystems and extreme weather events.
- Continuous monitoring and early warning systems are essential for climate resilience.

RELEVANCE FOR UPSC

- GS-I: Geography** — ENSO, Walker circulation, ocean–atmosphere interactions, climatic phenomena
- GS-III: Agriculture** — impact on crops, food security, irrigation, disaster management
- GS-III: Environment** — marine ecosystems, coral reefs, climate change, extreme weather
- GS-III: Indian Economy** — food inflation, agriculture, climate risks, resilient infrastructure

Also useful for: Essays (climate change & food security), Disaster management, Sustainable development

OCEANS CHANGE. SO DOES OUR TOMORROW.
UNDERSTANDING ENSO BUILDS A CLIMATE-RESILIENT INDIA.



KNOW TODAY
PREPARE FOR TOMORROW

BELOW-NORMAL SEPTEMBER RAINFALL RAISES CROP CONCERNS

Rainfall patterns are becoming crucial for agricultural output after an uneven monsoon, making the link between the southwest monsoon, kharif crops and food inflation important.

Good monsoon supports farm output, rural incomes and controls food inflation.
Deficient rainfall can reduce crop yields, strain reservoirs and increase food prices.

RAIN TODAY
FOOD TOMORROW
FARMERS STRONGER
INDIA SAFER

MONSOON
MATTERS
FOR A FOOD-
SECURE INDIA

ALL-INDIA RAINFALL STATUS

(June – September 2026, up to early September)



Cumulative rainfall is below normal, with September showing a deficit.

**SEPTEMBER RAINFALL
BELOW NORMAL**

Period	Actual (mm)	Normal (mm)	% Departure
June	102.3	96.1	+6%
July	285.4	290.1	-2%
August	248.7	264.9	-6%
September (upto 3 Sep)	28.4	86.7	-67%
Season (Jun-Sep)	666.7	741.8	-10%

Note: Figures are illustrative and based on IMD updates (as of early September 2026).

HOW THE SOUTHWEST MONSOON WORKS

- Land-sea heating contrast**
Indian land heats faster than ocean in summer, creating low pressure over land and high pressure over ocean.
- Moisture-laden winds**
Southwest monsoon winds from the Indian Ocean bring moisture towards India.
- Orographic uplift**
Winds rise over Western Ghats, Himalayas and other ranges, causing rain.
- Branching**
Arabian Sea branch and Bay of Bengal branch cover different parts of India.
- Retreat**
Monsoon withdraws from northwest India by September.

FACTORS AFFECTING MONSOON RAINFALL

- ENSO (El Niño/La Niña)**
 - El Niño → weaker monsoon, below-normal rainfall
 - La Niña → stronger monsoon, above-normal rainfall
- Indian Ocean Dipole (IOD)**
 - Positive IOD → supports better monsoon rainfall
 - Negative IOD → can weaken monsoon
- Snow cover over Himalayas**
Less snow cover can weaken the monsoon.
- Land surface and heat waves**
Higher pre-monsoon heat can strengthen monsoon circulation (to an extent).
- Cyclonic systems**
Monsoon depressions and low-pressure systems enhance rainfall.
- Climate change**
Increasing variability, extreme rainfall events and longer dry spells.

MONSOON AND KHARIF CROPS

Kharif crops depend heavily on timely and well-distributed rainfall.



RICE

High dependence on monsoon



MAIZE

Moderate dependence



PULSES

(tur, urad, moong)

Sensitive to dry spells



OILSEEDS

(soybean, groundnut)

Affected by rainfall distribution



COTTON

Needs adequate but well-spread rainfall

RESERVOIR STORAGE STATUS

Uneven monsoon has led to lower reservoir levels in some regions.



~70%
of total reservoir capacity
(as of early September 2026)

Lower storage may affect irrigation and rabi sowing if the deficit persists.

IMPACT ON AGRICULTURE AND FOOD INFLATION

- Lower crop yields if rainfall deficit continues.
- Reduced farm incomes, especially for rainfed areas.
- Tighter supply of foodgrains, pulses and oilseeds.
- Upward pressure on food inflation (e.g., cereals, vegetables, pulses).
- Greater need for buffer stock releases and import management.

RECENT TRENDS IN FOOD INFLATION

Food inflation remains sensitive to monsoon performance.



A prolonged rainfall deficit can push food inflation higher, affecting overall inflation and household welfare.

KEY RISKS TO WATCH

- Continued below-normal September rainfall
- Impact on kharif sowing and yields
- Decline in reservoir levels
- Spike in food prices (cereals, pulses, vegetables, edible oils)
- Possibility of El Niño strengthening
- Higher vulnerability of rainfed regions and small farmers

POLICY RESPONSE OPTIONS

- Accelerate irrigation support and micro-irrigation schemes
- Ensure adequate buffer stocks (FCI)
- Calibrated imports to manage prices if needed
- Crop insurance and direct income support
- Invest in climate-resilient agriculture and drought-tolerant varieties
- Strengthen weather forecasting and advisories for farmers

RELEVANCE FOR UPSC

- GS-I: Indian Geography** – Monsoon mechanism, ENSO, IOD, climatic factors
- GS-III: Agriculture** – Cropping patterns, food security, irrigation, reservoir management
- GS-III: Indian Economy** – Food inflation and its impact, government response
- GS-III: Environment & Climate Change** – Increasing variability and extreme events

KEY TAKEAWAYS

- ✓ September rainfall is below normal, raising crop concerns.
- ✓ Monsoon performance directly affects kharif output and food inflation.
- ✓ ENSO, IOD and other factors influence monsoon strength.
- ✓ Need for climate-resilient agriculture and effective policy response.



**STABLE
MONSOON
STRONG AGRICULTURE
FOOD SECURITY
PROSPEROUS INDIA**

INDIA'S STRONG GDP GROWTH SPARKS A MEASUREMENT AND DEVELOPMENT DEBATE

High growth numbers, but are jobs and investment keeping pace?

India's GDP growth remains strong, yet questions are being raised on whether the headline numbers are translating into sufficient employment, private investment and inclusive development.

HIGH GROWTH. BIGGER OPPORTUNITIES. INCLUSIVE DEVELOPMENT REMAINS THE GOAL.

SUSTAINABLE GROWTH FOR A DEVELOPED INDIA

LATEST GDP GROWTH

India's real GDP grew by
7.8%
in FY 2025-26
(as per latest official estimates)

Among the fastest-growing major economies in the world.

BUT KEY QUESTIONS REMAIN



Is enough quality employment being created?



Is private investment reviving strongly?



Are the benefits of growth reaching all sections of society?



Is the growth path sustainable and inclusive?

INDIA IN GLOBAL CONTEXT

Real GDP Growth (Latest available, %)



GDP vs GVA — WHAT'S THE DIFFERENCE?

Aspect	GDP (Gross Domestic Product)	GVA (Gross Value Added)
Meaning	Total value of final goods and services produced in an economy.	Value added at each stage of production (output minus intermediate consumption).
Measurement approach	Product approach, income approach, or expenditure approach.	Sum of value added across sectors (agriculture, industry, services).
Includes	Adds net indirect taxes (subsidies subtracted).	Does not include indirect taxes; focuses on sectoral value addition.
Relationship	$GDP = GVA + (\text{Net indirect taxes})$	
Use	Overall size of the economy.	Understand sectoral contribution to growth.

GROWTH, EMPLOYMENT AND INVESTMENT TRENDS

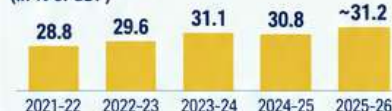
Real GDP Growth (%)



Unemployment Rate (%)



Gross Fixed Capital Formation (in % of GDP)



WHY THE DEBATE?



Employment not keeping pace with output growth.



Private investment remains below pre-pandemic levels.



Consumption is strong, but dominated by government spending in recent years.



Concerns over income and regional inequalities.



Need to translate growth into human development outcomes (health, education, skilling).

FACTORS BEHIND STRONG GDP GROWTH



Robust services sector (finance, IT, communications)



Higher public capital expenditure on infrastructure



Resilient domestic consumption



Strong exports in key sectors (pharma, electronics, engineering)



Macroeconomic stability and structural reforms

CHALLENGES AND CONCERNS



Job creation is less than expected, especially in formal manufacturing.



Private investment is subdued due to global uncertainty and domestic demand-supply gaps.



Quality of jobs and low female labour force participation.



Uneven regional development across states.



Risk of jobless growth if the manufacturing sector does not expand sufficiently.

THE WAY FORWARD



Revive private investment through stable policies, ease of doing business and lower regulatory uncertainty.



Focus on labour-intensive manufacturing and MSMEs.



Invest in skilling, education and health to improve productivity.



Promote inclusive growth with regional balance.



Leverage global opportunities while building domestic resilience.

RELEVANCE FOR UPSC

GS-III: Indian Economy

- Measurement of national income (GDP, GVA)
- Investment, employment and inclusive growth
- Issues related to structural transformation and jobless growth
- Policies for sustainable and balanced development

KEY TAKEAWAYS

- ✓ India's GDP growth remains strong.
- ✓ Debate exists on whether growth is translating into adequate employment and investment.
- ✓ Need to focus on quality, inclusive and sustainable growth.
- ✓ Both GDP and GVA are important for understanding economic performance.

“Growth is meaningful only when it creates opportunities for all.”

Towards a Developed and Inclusive India

RBI MOVES TO ABSORB RECORD BANKING LIQUIDITY

RBI announces ₹7 trillion, 30-day Variable Rate Reverse Repo (VRRR) auction with early-redemption facility

Move comes against a banking-system liquidity surplus of around ₹10.3 trillion.

STABLE
LIQUIDITY
STRONGER
ECONOMY

PRUDENT
POLICY
FOR A RESILIENT
INDIA

KEY NUMBERS



₹7 trillion
VRRR auction size
(30-day tenor)



₹10.3 trillion
Banking-system
liquidity surplus
(approx.)



30 days
Tenor of VRRR



Early exit
Early-redemption
facility announced

“ The operation aims to absorb surplus liquidity and ensure orderly conditions in the money market.

— Reserve Bank of India

WHY IS BANKING-SYSTEM LIQUIDITY HIGH?



High government spending
(capex and revenue outlay)



RBI's dividend transfer to the
government



Higher currency in circulation
(e.g., festive demand)



Moderate credit demand



Capital inflows and other durable
liquidity sources

WHAT IS VRRR?



Variable Rate Reverse Repo (VRRR)

- A monetary policy tool used by RBI to absorb short-term surplus liquidity from the banking system.
- Banks park their surplus funds with RBI at a variable interest rate (determined through auction).
- Funds are withdrawn from the system for a specified period (here 30 days).
- This auction has an early-redemption facility, allowing banks to exit before maturity if needed.

REPO vs REVERSE REPO vs VRRR

Feature	Repo	Reverse Repo	VRRR
Who borrows?	Banks borrow from RBI	Banks lend to RBI	Banks lend to RBI (auction)
Purpose	Injects liquidity	Absorbs liquidity	Absorbs large surplus liquidity
Rate	Fixed (repo rate)	Fixed (reverse repo rate)	Variable (auction-based)
Tenor	Usually overnight/short	Usually overnight/short	Specified (30 days in this case)
Flexibility	Regular tool	Regular tool	Used for large, transient liquidity
Current use	Not applicable	Limited use now	₹7 trillion, 30-day VRRR with early exit

RBI'S LIQUIDITY MANAGEMENT TOOLS



Repo and Reverse Repo
Adjusts short-term liquidity and policy rates



VRRR / Variable Rate Repo
Fine-tunes liquidity for longer/variable tenors



OMOs (Open Market Operations)
Buy/sell government securities



MSF (Marginal Standing Facility)
Provides overnight liquidity at a penal rate



Standing Deposit Facility (SDF)
Absorbs surplus liquidity (replaced fixed reverse repo)



CRR / SLR
Structural liquidity management tools



FX operations
Manage rupee liquidity through dollar buying/selling

IMPACT OF THIS MOVE



Helps keep short-term
interest rates aligned
with the policy stance



Prevents excessive
liquidity from pushing
down money-market rates



Maintains financial stability
and orderly market functioning



Supports monetary policy
transmission



Provides flexibility with
early exit option

WHY IT MATTERS?

- 1 Ensures effective control over system liquidity.
- 2 Keeps short-term interest rates close to the policy corridor.
- 3 Helps anchor inflation by supporting monetary policy transmission.
- 4 Prevents asset price distortions due to excess liquidity.
- 5 Signals RBI's proactive and flexible liquidity management.

KEY TAKEAWAYS FOR UPSC

- ✓ VRRR is used to absorb surplus liquidity from the banking system.
- ✓ It allows RBI to manage short-term liquidity without changing the policy rate.
- ✓ Repo injects liquidity; reverse repo and VRRR absorb liquidity.
- ✓ RBI uses multiple tools: repo, reverse repo, VRRR, OMOs, MSF, SDF, CRR, SLR, and FX operations.
- ✓ Liquidity management helps in monetary policy transmission, inflation control and financial stability.
- ✓ Recent move: ₹7 trillion, 30-day VRRR auction with early-redemption facility, against a ₹10.3 trillion banking-system liquidity surplus.

RELEVANCE FOR UPSC



GS-III: Indian Economy

- Monetary policy and its tools
- Money and banking
- Financial markets
- Macroeconomic stability
- Inflation and growth



Also useful for:

- **Prelims:** Definitions, tools, current data
- **Mains:** Role of RBI in ensuring financial stability, transmission of monetary policy



**SOUND LIQUIDITY MANAGEMENT
FOR A STRONGER, STABLE AND INCLUSIVE INDIA**

**STABLE MARKETS
STRONGER TOMORROW**

INDIA'S FOREIGN EXCHANGE RESERVES HIT FRESH RECORD

RBI data show forex reserves at **\$740.8 billion**
(for the week ended August 28, 2026)

Stronger dollar inflows, robust external sector and valuation gains have pushed India's reserves to a new high.

A STRONGER
RUPEE
A MORE
RESILIENT
INDIA



Forex Reserves
\$740.8 billion
(as on August 28, 2026)
▲ New All-Time High



Increase
+\$6.6 billion
(over previous week)

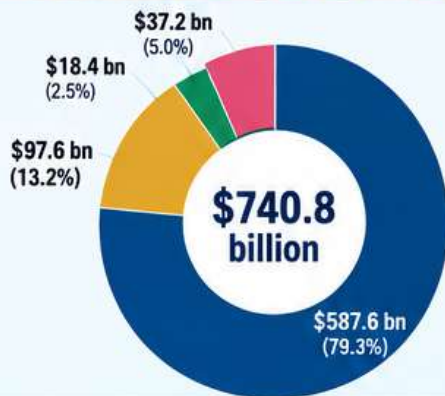


Previous Week
\$734.2 billion
(as on August 21, 2026)



India's Forex Reserves among the
LARGEST IN THE WORLD
(4th largest, after China, Japan, Switzerland)

COMPONENTS OF INDIA'S FOREX RESERVES (As on August 28, 2026)



- Foreign Currency Assets (FCA)**
Assets in major foreign currencies (e.g., USD, EUR, GBP, JPY)
- Gold Reserves**
Monetary gold held by RBI
- Special Drawing Rights (SDRs)**
Reserve asset created by the IMF
- IMF Reserve Position**
India's reserve position with the IMF

WHAT ARE FOREX RESERVES?



Foreign exchange reserves are external assets held by the RBI in foreign currencies, gold, SDRs and IMF reserve position. They act as a buffer against external shocks and help maintain stability in the exchange rate.

“Adequate forex reserves strengthen macroeconomic stability, build investor confidence and help the economy withstand global uncertainties.”

– Reserve Bank of India

WHY DO FOREX RESERVES MATTER?



Stabilise the rupee
Help RBI intervene in forex markets to prevent excessive volatility.



Meet external obligations
Enable payment for imports, external debt, and other international commitments.



Build investor confidence
A strong reserve position signals economic strength and creditworthiness.



Act as a buffer during crises
Useful during global financial shocks, commodity price spikes or capital outflows.



Support macroeconomic stability
Helps maintain low inflation and stable growth.

HOW ARE RESERVES BUILT?



Export earnings
Receipts in foreign currency



Capital inflows
FDI, FPI and external borrowings



RBI intervention
Purchase of foreign currency in the market



Valuation gains
Appreciation of non-USD assets (e.g., gold, euro, yen)

RBI AND EXCHANGE-RATE MANAGEMENT



Managed float
RBI intervenes in the market to smoothen volatility, not to maintain a fixed rate.



Tools used

- Buying/selling dollars
- Currency swaps
- Forward contracts
- Moral suasion and market communication



Objective
Ensure orderly movement in the rupee and maintain overall macroeconomic and financial stability.

INDIA IN GLOBAL CONTEXT

Forex Reserves (US\$ billion, latest available)



KEY FACTS FOR UPSC

- ✓ Foreign Currency Assets include government securities, treasury bills and other liquid assets.
- ✓ Gold reserves are valued at market prices.
- ✓ SDRs are international reserve assets created by the IMF in 1969.
- ✓ IMF reserve position reflects India's quota contribution and associated rights.
- ✓ RBI follows a managed floating exchange-rate regime.
- ✓ Adequate reserves are assessed using metrics like import cover, short-term external debt and composite adequacy frameworks (e.g., IMF ARA metric).

RELEVANCE FOR UPSC

- ✓ **GS-III: Indian Economy**
 - Balance of Payments
 - Exchange rate management
 - External sector and macroeconomic stability
- ✓ **Related topics:** IMF, SDRs, global financial architecture
- ✓ **Link with current affairs:** Global economic uncertainty, capital flows, dollar strength, commodity prices
- ✓ **Also relevant for:** Indian economic resilience and policy responses

A RESILIENT INDIA
IN A VOLATILE WORLD

STRONGER
EXTERNAL SECTOR
BRIGHTER TOMORROW



STABILITY



CONFIDENCE



OPPORTUNITIES



PROSPERITY

2026-09-06

DAILY EDITION

READ
LEARN
STAY INFORMED
GROW

IMPORTANT DEVELOPMENTS. BROADER PERSPECTIVES.

CURRENT AFFAIRS

DAILY DIGEST

NEWS THAT SHAPES A BETTER TOMORROW



POLITY &
GOVERNANCE



ECONOMY



INTERNATIONAL
RELATIONS



ENVIRONMENT



SCIENCE &
TECHNOLOGY



AGRICULTURE

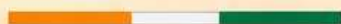


GEOGRAPHY



POLICIES &
SCHEMES

A MORE INFORMED
INDIA FOR A BRIGHTER
TOMORROW



PEOPLE
PLANET
PROGRESS



INNOVATION
FOR A STRONGER
TOMORROW



A MORE
PROSPEROUS
INDIA



SUSTAINABLE
GROWTH



OUR LAND
OUR HERITAGE



A HEALTHIER
PLANET

SAME FACTS. A BRIGHTER TOMORROW.

INDIA'S VULTURE CONSERVATION FROM CRISIS TO RECOVERY

Healthy Ecosystems | Safer Communities | A Vulture-friendly India

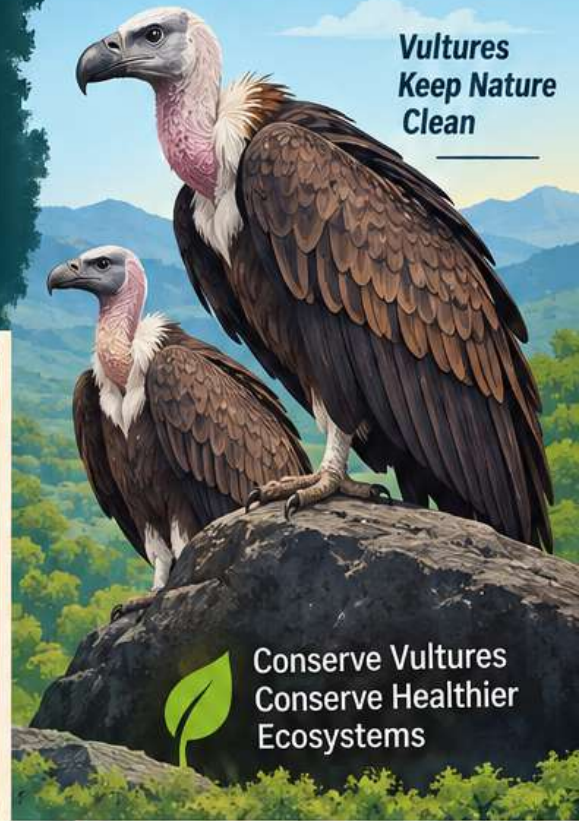
Vultures
Keep Nature
Clean

WHAT'S NEW?

6 September 2026



- Union Environment Minister inaugurated a national workshop at **Panchkula, Haryana** for **International Vulture Awareness Day 2026**.
- Workshop theme: **'From Crisis to Recovery – Conserving India's Vultures'**.
- Call for continued creation of **Vulture Safe Zones** and stronger technical capacity for species recovery in the wild.
- **Organised by:** Ministry of Environment, Forest and Climate Change, Central Zoo Authority, Haryana Forest & Wildlife Department, Bombay Natural History Society, and Jatayu Conservation Breeding Centre at Pinjore.



Conserve Vultures
Conserve Healthier
Ecosystems

WHY VULTURES MATTER



Scavengers rapidly dispose of carcasses.



Reduce disease risk for humans and livestock.



Help in nutrient cycling.



Indicator of ecosystem health.

WHY POPULATIONS CRASHED



Veterinary **diclofenac poisoning** through livestock carcasses was a major driver.



Habitat and food-chain pressures.



Unsafe carcass management and other threats.

VULTURE SAFE ZONE



- A landscape where threats are reduced, especially toxic veterinary drugs.
- Ensures safe food sources and proper carcass management.
- Involves monitoring, community participation and veterinary engagement.

INDIA FACTS

9

India has **nine vulture species**.

Examples:

- White-rumped Vulture
- Indian (Long-billed) Vulture
- Slender-billed Vulture
- Red-headed Vulture
- Egyptian Vulture
- Himalayan Griffon



CONSERVATION TOOLKIT



Captive breeding and reintroduction.



Drug safety and veterinary awareness (e.g. ban on diclofenac).



Habitat protection and safe carcass management.



Population monitoring and research.



Community participation and multi-stakeholder action.

PINJORE CENTRE



Jatayu Conservation Breeding Centre (JCBC) at Pinjore, Haryana

- A major conservation-breeding facility for vultures.
- In 2025–26, it housed **356 vultures** and bred **35 chicks**.



PRELIMS BOX



Vultures are birds of prey (raptors) and **scavengers**, not predators of live prey in normal circumstances.



Diclofenac residues in cattle carcasses can cause fatal kidney failure in vultures.



In-situ: Vulture Safe Zones (in the wild).
Ex-situ: Conservation breeding in facilities like JCBC, Pinjore.

Vultures Today
Healthier Tomorrows

SAVE VULTURES
SAVE ECOSYSTEMS
SAVE LIVES

INDIA – SRI LANKA

DEFENCE & MARITIME PARTNERSHIP

Stronger Neighbours, A Safer Indian Ocean

PEACE
STABILITY
PROSPERITY
IN THE INDIAN OCEAN

WHAT'S NEW?



Rajnath Singh visit:
8 – 10 Sept 2026



High-level India delegation

Including officials from Defence and External Affairs



Focus: defence + maritime cooperation

Will hold consultations with Sri Lankan leadership and interact with the Indian diaspora in Colombo.

INDIA AND SRI LANKA: KEY MARITIME GEOGRAPHY



WHY SRI LANKA MATTERS



India's closest southern maritime neighbour



Located beside key Indian Ocean sea lanes



Palk Strait & Gulf of Mannar link the two countries



Important for regional security and Indian Ocean stability

DEFENCE COOPERATION



SLINEX: India–Sri Lanka bilateral naval exercise



MITRA SHAKTI: bilateral army exercise



IN – SLN DIVEX: diving & submarine-rescue cooperation



HADR and maritime interoperability

STRATEGIC CONTEXT



Neighbourhood First



Security and Growth for All in the Region (SAGAR)



Maritime domain awareness



Countering non-traditional security threats

PRELIMS FACTS



Sri Lanka lies south of India



Palk Strait separates Tamil Nadu and Sri Lanka



Gulf of Mannar is between southeast India and western Sri Lanka



Colombo is a major Indian Ocean port

A STRONGER PARTNERSHIP
FOR A STABLE AND PROSPEROUS INDIAN OCEAN



INDIA-JAPAN

ELEVATE TIES TO A NEW STRATEGIC PHASE

During the recent visit of Japan's Prime Minister Shigeru Ishiba to India (5-6 Sept 2026), the two countries launched a new 10-year roadmap to deepen the India-Japan Special Strategic and Global Partnership.

STRONGER
PEOPLE • PROSPERITY
• A STABLE INDO-PACIFIC



'VISION FOR THE NEXT DECADE'

A forward-looking plan to make India-Japan cooperation more ambitious, people-centric and impactful, anchored in shared democratic values.



PEOPLE-TO-PEOPLE & HUMAN RESOURCE MOBILITY

- Target to facilitate **500,000 people-to-people exchanges** in the next decade.
- Greater opportunities for students, skilled professionals and cultural exchanges.



ECONOMIC & INVESTMENT TIES

- Aim to invest **₹10 lakh crore** in India over the next 10 years.
- Focus on infrastructure, manufacturing, clean energy, semiconductors and startups.
- Stronger collaboration for resilient and diversified supply chains.



GREEN GROWTH & SUSTAINABILITY

- Enhanced cooperation in clean energy, energy transition and climate action.
- Collaboration on green hydrogen, battery technologies and sustainable infrastructure.
- Support for a low-carbon, climate-resilient and inclusive growth model.



DEFENCE & SECURITY COOPERATION

- Further strengthen defence and strategic ties.
- Greater collaboration in maritime security, a free, open and rules-based **Indo-Pacific**.
- Enhanced defence industrial cooperation and technology exchange.



TECHNOLOGY & INNOVATION

- Deeper cooperation in AI, semiconductors, quantum technologies, space and next-generation technologies.
- Joint research, innovation ecosystems and startup linkages.



REGIONAL & GLOBAL COOPERATION

- Reaffirm commitment to a free, open, inclusive and prosperous **Indo-Pacific**.
- Closer coordination in multilateral forums (**Quad, G20, UN**, etc.).
- Shared efforts for peace, stability and a rules-based international order.



TWO DEMOCRACIES. A STRONGER TOMORROW.

INDIA - JAPAN



GLOBAL BIOFUELS ALLIANCE

CLEAN ENERGY. STRONGER TOMORROW.

A global partnership led by India to accelerate the adoption of sustainable biofuels.



WHAT IS IT?



An international initiative launched by **India** in 2023 during the G20 presidency.



Aims to create a global ecosystem for production, use and trade of sustainable biofuels (bioethanol, biodiesel, SAF, biogas, etc.).



Brings together governments, international organisations, industry and research institutions.

MEMBERS

12 MEMBER COUNTRIES



India



USA



Brazil



Argentina



Canada



Italy



Mauritius



South Africa



UAE



Singapore



Bangladesh

and
1 more

INTERNATIONAL ORGANISATIONS



IEA

International Energy Agency



IRENA

International Renewable Energy Agency



FAO

Food and Agriculture Organization

KEY FOCUS AREAS



Scale up production and use of biofuels (ethanol, biodiesel, SAF, biogas)



Facilitate technology development and knowledge sharing



Promote sustainable and resilient supply chains



Mobilise finance and investment



Develop common standards and certification



Support policy cooperation and capacity building

WHY BIOFUELS?



Lower greenhouse gas emissions



Cleaner air and better public health



Reduces dependence on imported fossil fuels



Helps rural economy and farmers (use of agri-residues like sugarcane, rice straw, etc.)

INDIA'S RELATED INITIATIVES



Ethanol Blending Programme (20% by 2025–26)



Sustainable Aviation Fuel (SAF) roadmap



Waste to Wealth Mission



National Bioenergy Programme



2G ethanol and advanced biofuel research

CHALLENGES



Sustainable and consistent feedstock supply



Higher production costs compared to fossil fuels



Need for advanced technology and infrastructure



Harmonising global standards



Balancing food security with biofuel expansion

FROM FIELDS TO A GREENER FUTURE

**BIOFUELS
DRIVE ENERGY SECURITY,
RURAL PROSPERITY AND
A CLEANER PLANET.**



SUSTAINABLE
ENERGY



STRONGER
FARMERS



CLEANER
ENVIRONMENT



A HEALTHIER
PLANET

INDIA LAUNCHES

MAITRI-II

A NEW ERA FOR INDIA'S POLAR RESEARCH

India's second indigenously built polar research vessel, MAITRI-II, has been launched, marking a major step in strengthening India's presence in the Antarctic and the Southern Ocean.



ABOUT MAITRI-II



Type: Polar Research Vessel (ice-class)



Built by: Garden Reach Shipbuilders & Engineers (GRSE), Kolkata



Indigenously designed and built



Launched: September 2026



Purpose: Support India's Antarctic missions, scientific research and logistics supply

KEY FEATURES



Ice-breaking capability

Can navigate through sea ice in polar regions



Accommodation

For ~100 scientists, crew and support staff



Advanced laboratories

For oceanography, climate, marine biology and earth sciences



Helideck

Supports helicopter operations for logistics and research



Cargo capacity

Carries fuel, food and equipment for long-duration missions



Environment-friendly

Designed with low emissions and complies with modern environmental standards

INDIA'S ANTARCTIC PROGRAMME



Began in 1981 with the first Indian Antarctic Expedition



Research stations:

Maitri (1989) and Bharati (2012)



Focus areas: climate change, glaciology, oceanography, atmospheric sciences, biology and geology



Operated by: National Centre for Polar and Ocean Research (NCPOR), Ministry of Earth Sciences

WHY IT MATTERS



- Strengthens India's scientific research in the Antarctic and Southern Ocean



- Helps in better understanding climate change and its impact on India's monsoon and coastal regions



- Supports strategic presence in a resource-rich and geopolitically important region



- Enhances India's role in global polar governance and scientific collaboration



- Facilitates study of oceans, biodiversity and extreme environments

THE ANTARCTIC TREATY



- Signed in 1959, in force since 1961
- Designates Antarctica for peaceful purposes and scientific research
- Prohibits military activity, nuclear explosions and disposal of nuclear waste
- India is a Consultative Party (since 1983) with voting rights

A CLEANER, SAFER PLANET



- Polar research helps monitor global climate change
- Provides early warning for sea-level rise and extreme weather
- Contributes to a sustainable and resilient future



FROM THE POLAR OCEAN TO A SAFER TOMORROW

SCIENCE | SUSTAINABILITY | A STRONGER INDIA



INDIA'S NATIONAL CRITICAL MINERAL MISSION (NCCM)

SECURING RESOURCES FOR A SELF-RELIANT AND
CLEAN ENERGY FUTURE

CRITICAL
MINERALS
STRONGER INDIA

LITHIUM
COBALT
NICKEL
RARE EARTHS
COPPER
GRAPHITE
AND MORE

WHAT IS NCCM?



A flagship mission by the Government of India to ensure long-term and sustainable availability of **critical minerals** for India's **economic growth**, **clean energy transition** and **strategic autonomy**.

LAUNCHED



**JANUARY
2025**

Total outlay:

₹34,300 CRORE
(2025–2031)

WHY IT MATTERS?



Supports **clean energy** (renewable energy, EVs, energy storage)



Reduces **import dependence**



Enhances **strategic and economic security**



Enables **high-technology manufacturing**



Essential for India's **net-zero climate goal**

KEY OBJECTIVES



Increase domestic exploration and mining of critical minerals



Secure overseas mineral assets



Develop processing and recycling capabilities in India



Build a skilled workforce and strong R&D ecosystem

CRITICAL MINERALS INCLUDE



Lithium
(EV batteries)



Cobalt
(Batteries, superalloys)



Nickel
(Batteries, stainless steel)



Rare Earth Elements
(Magnets, electronics)



Copper
(Clean energy, electricals)



Graphite
(Batteries, nuclear)

MAJOR COMPONENTS



Geological mapping and mineral exploration



Acquisition of overseas mines and strategic partnerships



Setting up processing, refining and value addition facilities



Promoting recycling and urban mining



Research, innovation and skilling initiatives

RECENT DEVELOPMENTS



First set of critical mineral blocks identified for auction in India



Discussions underway for overseas assets in resource-rich countries (Australia, Africa, Latin America)



Focus on building domestic processing and refining capacity



Industry and global partners showing strong interest

THE BIGGER PICTURE

CRITICAL MINERALS POWER
THE TECHNOLOGIES OF TOMORROW –
**CLEANER PLANET,
STRONGER ECONOMY,
SECURE INDIA.**



RENEWABLE ENERGY



ELECTRIC VEHICLES



ELECTRONICS



DEFENCE



GREEN HYDROGEN



INDIA - EU

FREE TRADE AGREEMENT

A STRONGER PARTNERSHIP
FOR A MORE PROSPEROUS FUTURE

TRADE
PEOPLE
PLANET
PROSPERITY

STRONGER
INDIA
STRONGER
EUROPE
A MORE STABLE
WORLD

LATEST DEVELOPMENT



Negotiations for the
India-EU Free Trade Agreement
(FTA) concluded in
September 2026.



The agreement is now moving
towards **legal review, translation
and formal approval.**

KEY FEATURES (REPORTED)



Reduction or elimination of tariffs
on majority of traded goods



Better market access for services
(IT, finance, professional services, etc.)



Facilitation of investments and
easier regulatory processes



Cooperation in technology, innovation
and digital trade



Focus on sustainable trade, clean energy
and critical raw materials



Provisions for labour rights, environmental
standards and inclusive growth

INDIA-EU TRADE: AT A GLANCE



EU is India's
largest trading partner.



Total trade (2024-25):
~€137 billion
(≈ ₹12.5 lakh crore)



India's exports to EU: **~€75 billion**



India's imports from EU: **~€62 billion**



EU accounts for **~17%** of India's
total trade.

WHAT IS IT?



A comprehensive trade agreement
to reduce tariffs, remove trade barriers
and deepen economic cooperation
between India and the European Union.

WHY IT MATTERS?



Creates new opportunities for Indian goods,
services and investments in the EU market.



Strengthens strategic partnership
democracies.



Supports resilient and diversified
global supply chains.



Helps in green transition, clean energy
and sustainable development.

KEY SECTORS THAT MAY BENEFIT



Textiles, leather, marine products



Automobiles and auto components



Pharmaceuticals and chemicals



Electronics and electrical goods



Agriculture and processed food



IT and business services



Gems and jewellery



Renewable energy and green technologies

THE ROAD AHEAD



- Legal scrubbing and translation
(of the agreement text)
- Approval by the European Council,
European Parliament and India's cabinet
- Final signing and ratification
- Implementation and monitoring
mechanisms



DEEPER TIES
BRIGHTER OPPORTUNITIES

INDIA & EU
PARTNERS
IN PROGRESS